

AD9988/AD9986/AD9081/AD9082/AD9177/AD9207/AD9209 API Release Notes v1.7.0

Rev 1.7.0 Release on Oct 2024

Release Summary

This release will provide the following key fixes

- New PFILT configuration interface
- Identifying Sysref frequency limitations
- Extras allowing tx only and rx only use cases

Deliverables Summary

Deliverable	Description	Release ID	Updated
API Device Libraries	Ansi C ADI Device libraries	v1.7.0	Yes
Eval App	Embedded Reference Application in C targeting evaluation platform. Simple command line application to configure datapaths and bring up links	v1.7.0	Yes
Extras App	Expanded Reference Example application targeting evaluation platform for tone generation and data capture from MZ board user-space.	v1.7.0	Yes

API Device Libraries Updates

New Features, Interface Changes:

Summary of API Interface changes, refer to main API interface header file (adi_ad9082.h,adi_ad9081.h) for full details.

Changes listed in this section may require updates in end applications. Refer to eval app source code for example usage.

Update	Reason for Update	Source Code Reference	Required /Recommended Optional	Application Change
adi_adxxxx_device_startup_nco_test_mode	Expose jesd mode setting for nco mode to cover variability of interpolation settings	adi_inc/adi_adxxxx.h	Recommended	Call adi_adxxxx_device_startup_nco_test_mode function with added jesd_mode parameter instead of adi_adxxxx_device_startup_nco_test function which has hardcoded jesd mode
adi_adxxxx_adc_pfir_coeff_table_load_set	Streamline PFIR block configuration and support all modes	adi_inc/adi_adxxxx.h	Recommended	Call adi_adxxxx_adc_pfir_coeff_table_load_set to configure PFIR block according to Table 119 in UG-1578
adi_adxxxx_sync_sysref_frequency_set	Generate sysref frequency value for rx only and tx only use cases without failure. Checks for limiting cases.	adi_inc/adi_XXXX.h	Required, Breaking change	adi_adxxxx_sync_sysref_frequency_set has new dev_ref parameter requiring the device reference clock as an input

Bug Fixes/ Improvements

Summary of API bug fixes and improvements

Changes listed in this section do not require updates in the end application

Bug Fixes/ Improvement	API Updated	Reason for Update	Required/Recommended Optional
Power down unused dacs	adi_adxxxx_device_startup_tx_or_nco_test	Assumed power down at default but all dacs are powered on by default	Recommended
Changed eye_scan_data vector to signed int Started spo sweep from 0 Scaled y-axis of eye scan to recommended 4mV	adi_adxxxx_jesd_rx_qr_two_dim_eye_scan	Eye scan output graph had unequal scaling and missing data points	Required
Remove coefficient clear and xfer calls from config function so current configuration does not get cleared	adi_adxxxx_adc_pfir_config_set	This function could only be used for single real configuration because when trying to load multiple modes, the coefficients would clear, losing all previously loaded values	Required
Warn users using onboard PLL of deterministic latency limitations	adi_adxxxx_device_clk_pll_startup	Only certain clocking configurations can be used when running in Subclass 1 using the onboard PLL	Optional

Known Issues

Eval Application Updates

New Features

- Introduce new file to store app layer helper functions: adxxxx_app_helper.c
- Support AD7175 configuration in application and display power measurements

Bug Fixes/ Improvements

Component	Update	Reason for Update
AD9988/AD9986/AD9081 API	Integrated v1.7.0	
Example App	Added new use case 41	To demonstrate PFILT using FBW use case
Example App	app_test_eye_scan app_test_eye_scan_write_to_file Allocated eye_scan_data array to pass through eye_scan functionality	Cleaner formatting
Example App	Sysref mode set to default = SYSREF_NONE	Only configure SYSREF for subclass 1 use cases
Example App	Move loopback lane mapping block to beginning of app	To implement changed lane mapping throughout the program
Example App	Call adi_adxxxx_jesd_oneshot_sync for Use case 0	NCO mode implements oneshot sync to synchronize NCO timing

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Example App	<code>uc_settings.c</code> Added a JESD mode for JRX for NCO mode aka Use case 0	New NCO mode function interface requires a JESD mode to be passed in from <code>uc_settings.c</code>
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Known Issues

- On the Evaluation board the HMC7044 PLL and MxFE Device PLL loop filter settings may not be optimal for all use-case included in application. It may be required to call the use-case a second time if the PLL doesn't lock. Alternatively external clocks can be applied. End application should optimize the PLL settings as per the Datasheet recommendations.
- On the Evaluation board, some use cases (5 and 8) display link failure due to a RX path configuration issue. This has been observed mainly on AD9986-FMCB-EBZ. Future releases will support adjustment of use case settings to accommodate full MxFE family.
- The ADS9 FPGA Board cannot configure single pulse oneshot sync for subclass 1 at this time. To use subclass 1 with the ADS9 FPGA platform, use a continuous SYSREF signal during oneshot sync. Future releases will support additional SYSREF signal configurations when operating in SC1 on the ADS9 board.

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Extras App Updates

New Features

- Support for AD7175 configuration available

Bug Fixes/ Improvements

Bug Fixes/ Improvement	API Updated	Reason for Update	Required/Recommended Optional
Check for tx link vs rx link up status before proceeding through main application	main	Allow tx only and rx only use cases to operate	Recommended

Known Issues

AD9988/AD9986/AD9081/AD9082/AD9177/AD9207/AD9209 API Release Notes v1.6.1

Rev 1.6.1 Release on Dec 2023

Release Summary

This release will provide the following key fixes

- Reenable DAC blocks for NCO only test in evaluation app

Deliverables Summary

Deliverable	Description	Release ID	Updated
API Device Libraries	Ansi C ADI Device libraries	v1.6.1	Yes
Eval App	Embedded Reference Application in C targeting evaluation platform. Simple command line application to configure datapaths and bring up links	v1.6.1	Yes
Extras App	Expanded Reference Example application targeting evaluation platform for tone generation and data capture from MZ board user-space.	v1.6.1	Yes

API Device Libraries Updates

New Features, Interface Changes:

Summary of API Interface changes, refer to main API interface header file (adi_ad9082.h ,adi_ad9081.h) for full details.

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Bug Fixes/ Improvements

Summary of API bug fixes and improvements

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Bug Fixes/ Improvement	API Updated	Reason for Update	Required/Recommended Optional
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Known Issues

Eval Application Updates

New Features

- Introduce new file to store app layer helper functions: adxxxx_app_helper.c
- Support AD7175 configuration in application and display power measurements

Bug Fixes/ Improvements

Component	Update	Reason for Update
AD9988/AD9986/AD9081 API	Integrated v1.6.1	
Example App	Removed NCO test UC0 from case statement to power down DAC blocks	DAC blocks necessary for NCO test

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Known Issues

- On the Evaluation board the HMC7044 PLL and MxFE Device PLL loop filter settings may not be optimal for all use-case included in application. It may be required to call the use-case a second time if the PLL doesn't lock. Alternatively external clocks can be applied. End application should optimize the PLL settings as per the Datasheet recommendations.
- On the Evaluation board, some use cases (5 and 8) display link failure due to a RX path configuration issue. This has been observed mainly on AD9986-FMCB-EBZ. Future releases will support adjustment of use case settings to accommodate full MxFE family.
- The ADS9 FPGA Board cannot configure single pulse oneshot sync for subclass 1 at this time. To use subclass 1 with the ADS9 FPGA platform, use a continuous SYSREF signal during oneshot sync. Future releases will support additional SYSREF signal configurations when operating in SC1 on the ADS9 board.
- The ADS9 FPGA Board capture delay feature sets the delay to (1+x) LMFC/LEMC cycles for all delays > 0 where x is the user input value for delay. For delay = 0, the delay is set to 0 LMFC/LEMC cycles.

Extras App Updates

New Features

- Support for AD7175 configuration available

Bug Fixes/ Improvements

Bug Fixes/ Improvement	API Updated	Reason for Update	Required/Recommended Optional

Known Issues

AD9988/AD9986/AD9081/AD9082/AD9177/AD9207/AD9209 API Release Notes v1.6.0

Rev 1.6.0 Release on Nov 2023

Release Summary

This release will provide the following key fixes

- Power measurement API for FMCB boards with new AD7175 ADC
- Power savings measures on DAC for use cases not using all DAC cores
- Capture gating by transmit feature on ADS9

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Extras App	Expanded Reference Example application targeting evaluation platform for tone generation and data capture from MZ board user-space.	v1.6.0	Yes
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API Device Libraries Updates

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Update	Reason for Update	Source Code Reference	Required /Recommended Optional	Application Change
adi_adxxxx_dac_duc_sel_sideband_enable_set	Control bitfield using API to allow frequency inversion by selecting the lower sideband image	adi_inc/adi_adxxxx.h	Optional	Can call API to invert DAC output on main and channel datapaths

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adi_adxxxx_dac_dll_power_down	Power down DLL clocks, SWD clocks, and mushi decoders based on input DACs	adi_inc/adi_adxxxx.h	Recommended	Used for Rx Only Use cases for power saving measures
adi_adxxxx_dac_duc_main_dig_gain_set	Allow DAC digital gain configuration	adi_inc/adi_adxxxx.h	Recommended	Can set the 12 bit digital gain on DAC datapath

Bug Fixes/ Improvements

Summary of API bug fixes and improvements

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Bug Fixes/ Improvement	API Updated	Reason for Update	Required/Recommended Optional
Optimized analog input buffer settings for DC coupled configurations Set ADC cores based on die ID rather than user input	adi_adxxxx_adc_analog_input_buffer_set	To optimize buffer performance for DC coupled applications and minimize user error	Recommended
Disable coarse and fine NCOs when in full bandwidth mode	adi_adxxxx_adc_path_or_bypass_config	To further optimize power when NCO path is unused during full bandwidth configurations	Recommended